

YOLO Model Testing Report

Model:	yolov8m_finetuned_20251127
Model Size:	49.6 MB
Parameters:	25.9 M
FLOPs (640x640):	79.09 GFLOPs
Run Name:	yolov8m_finetuned_20251127_testing_20251127_034915
W&B Run Name:	yolov8m_finetuned_20251127_bdd100k_yolo_limited_test_20251127_034915
Timestamp:	2025-11-27 03:52:23
Dataset:	bdd100k_yolo_limited - test split
Images Processed:	20000
Total Execution Time	124.67s
IoU Threshold:	0.5

Inference Performance

Metric	Value
Average Inference Time	2.35 ms per image
FPS (Frames Per Second)	424.67

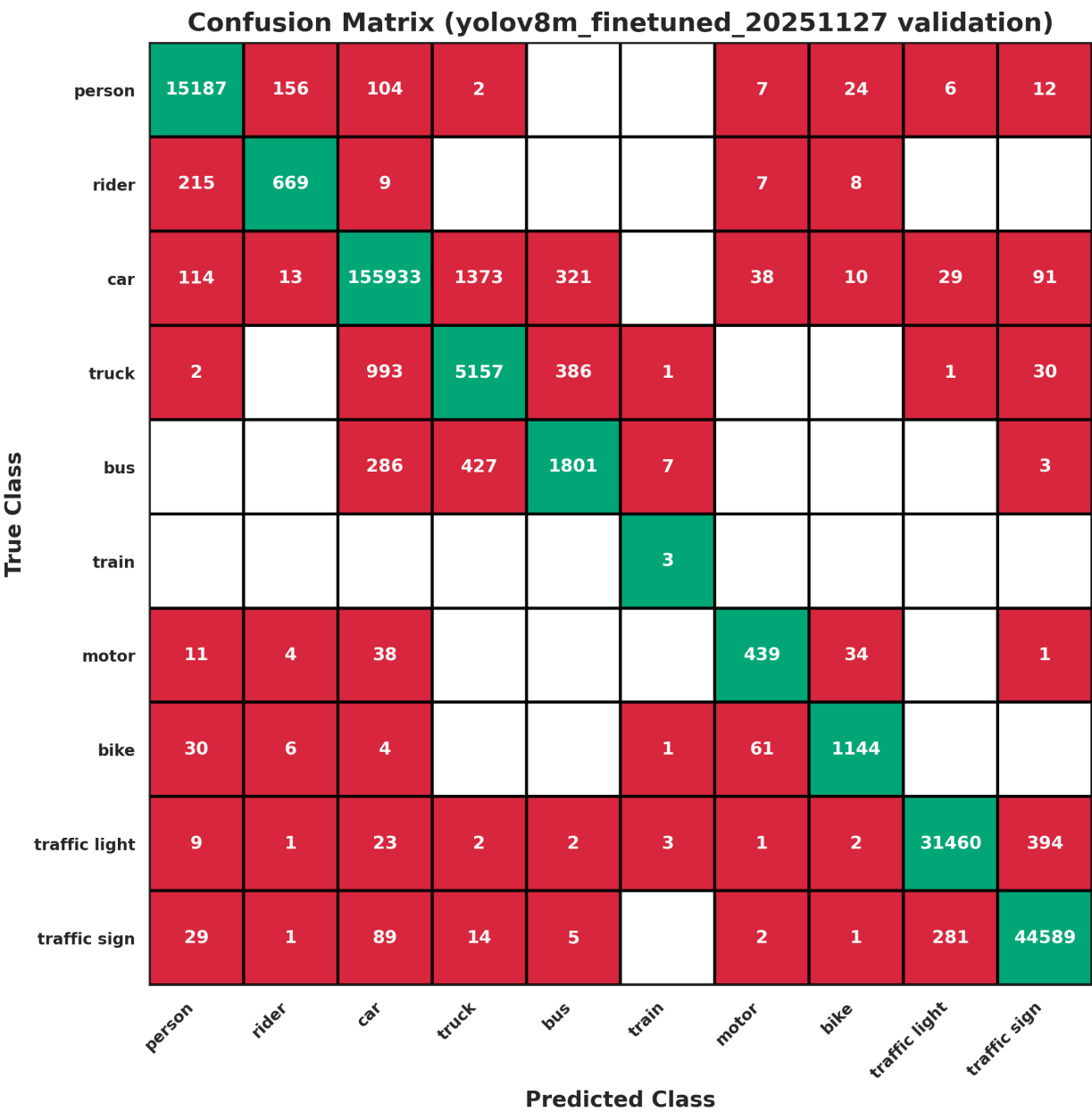
Note: Inference time includes preprocessing, model inference, and postprocessing.

Overall Accuracy Metrics

Metric	Value
Precision	0.6972
Recall	0.7814
F1-Score	0.7369
mAP@0.5	0.5683

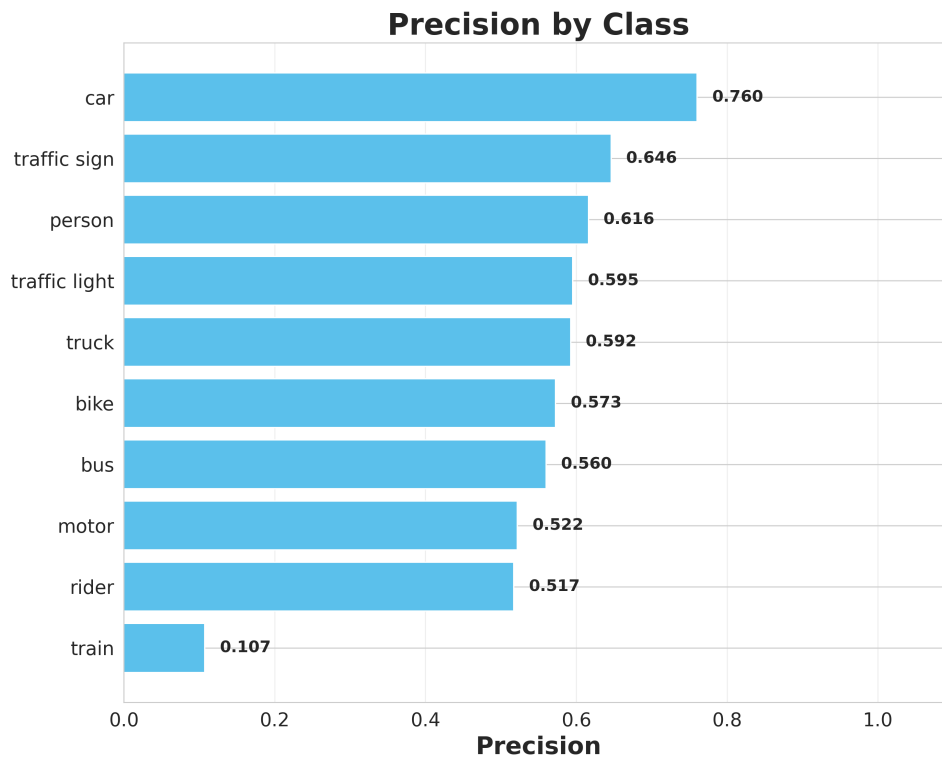
mAP@0.5:0.95	0.3236
True Positives	256382
False Positives	111345
False Negatives	71716

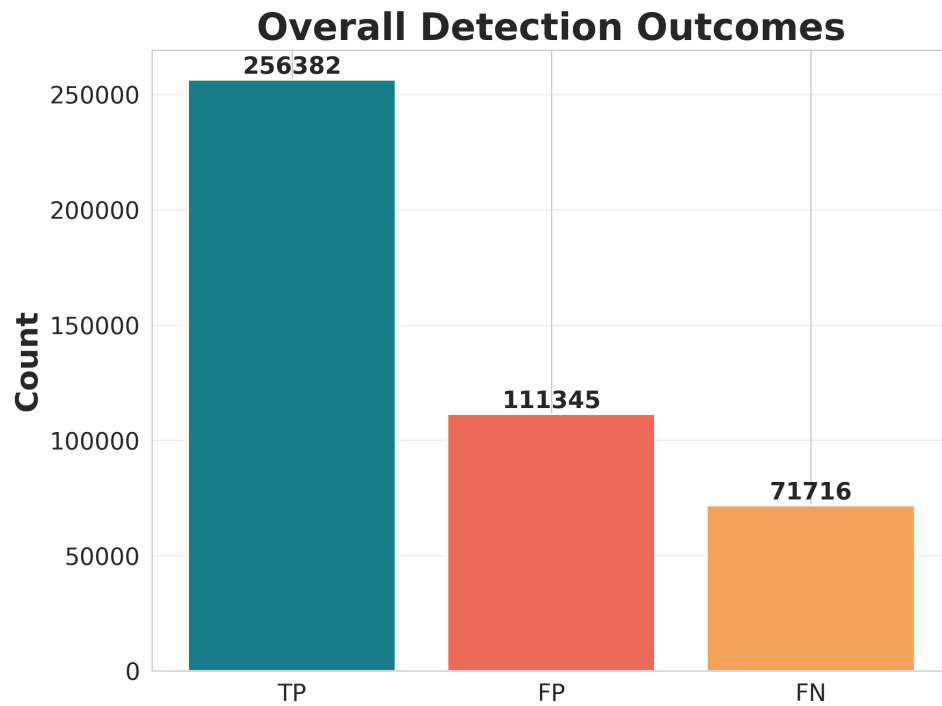
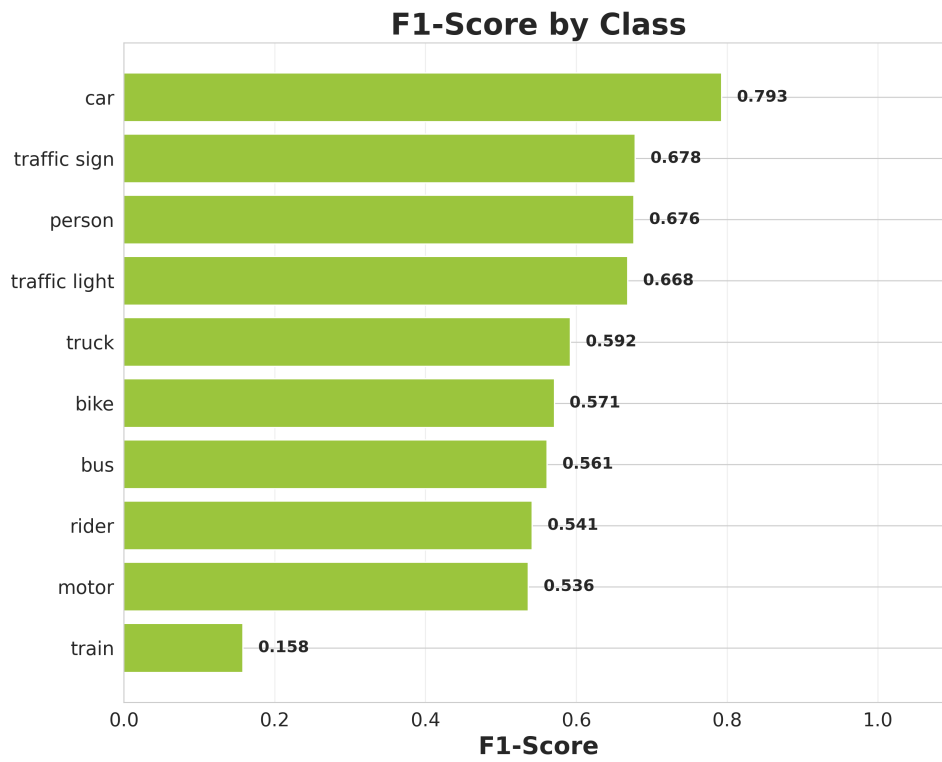
Confusion Matrix

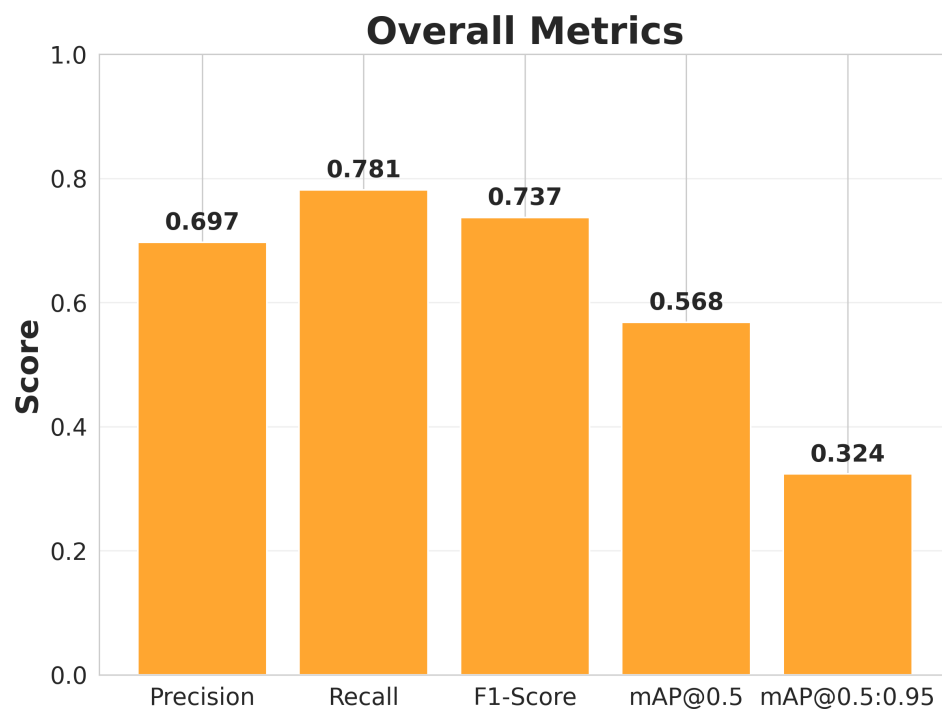
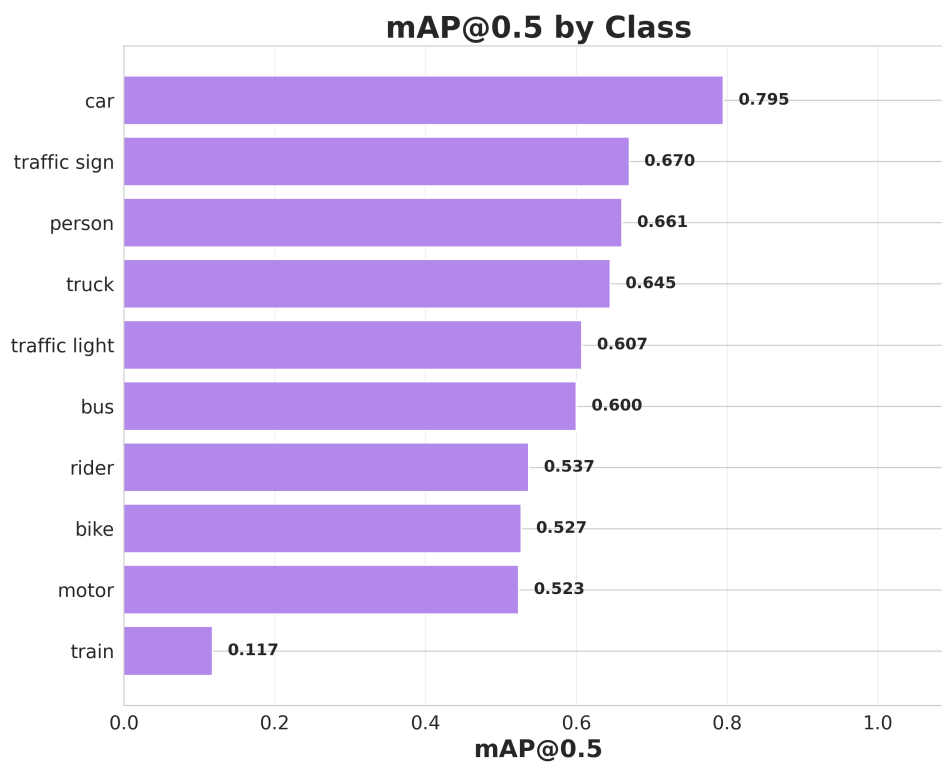


Correct Predictions (Diagonal Sum): 256382
Total Matched Predictions: 433719

Performance Visualizations







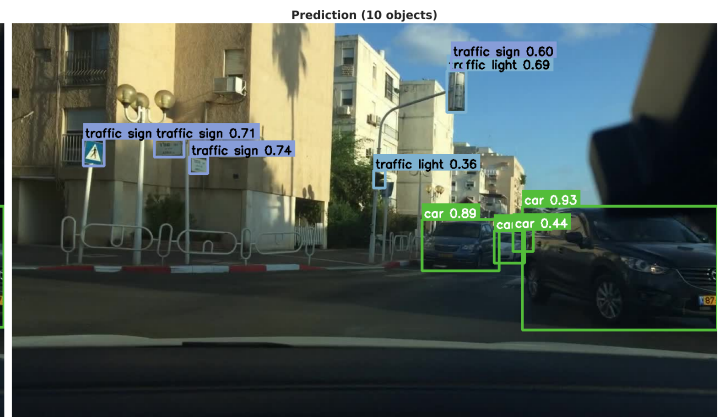
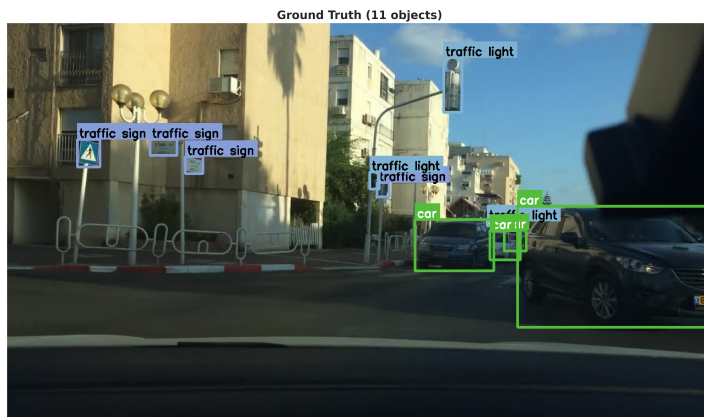
Per-Class Performance

Class	TP	FP	FN	Precision	Recall	F1-Score	mAP@0.5
person	15187	9463	5071	0.6161	0.7497	0.6764	0.6605
rider	669	625	508	0.5170	0.5684	0.5415	0.5368
car	155933	49216	32234	0.7601	0.8287	0.7929	0.7951
truck	5157	3547	3553	0.5925	0.5921	0.5923	0.6452
bus	1801	1416	1399	0.5598	0.5628	0.5613	0.5998
train	3	25	7	0.1071	0.3000	0.1579	0.1174
motor	439	402	357	0.5220	0.5515	0.5363	0.5235
bike	1144	854	865	0.5726	0.5694	0.5710	0.5267
traffic light	31460	21380	9822	0.5954	0.7621	0.6685	0.6072
traffic sign	44589	24417	17900	0.6462	0.7135	0.6782	0.6702

Sample Predictions: Ground Truth vs Model

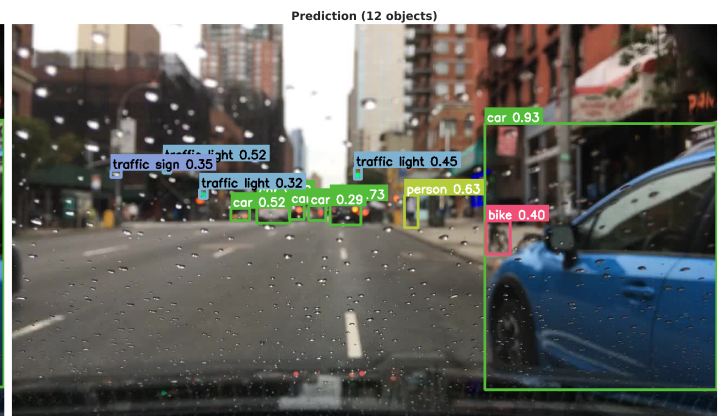
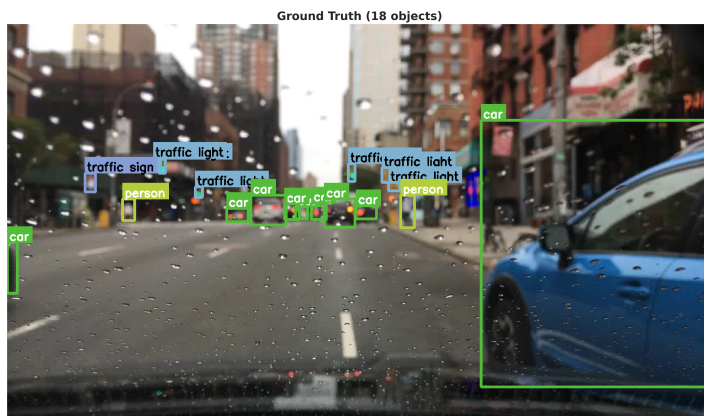
Sample #1: ccd007e9-bded7ebd.jpg

Ground Truth Objects: 11 | Predicted Objects: 10



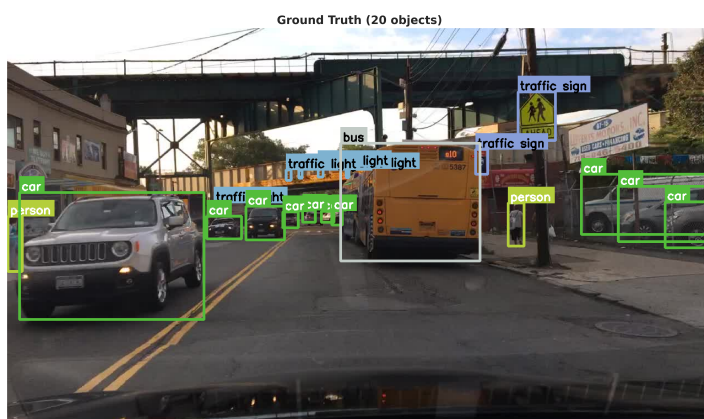
Sample #2: f9abbc64-92583c3d.jpg

Ground Truth Objects: 18 | Predicted Objects: 12



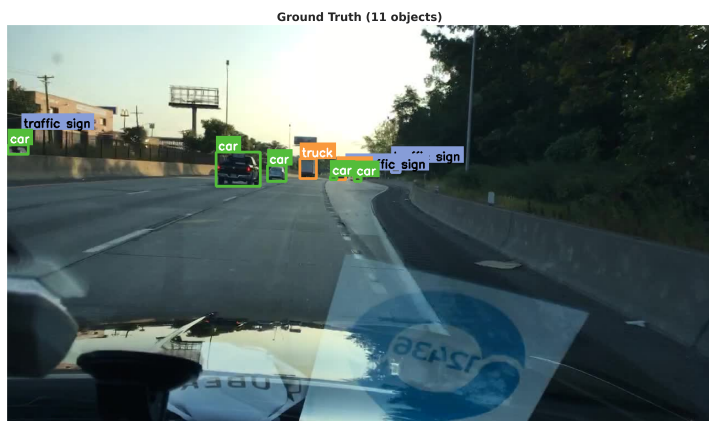
Sample #3: db1385a4-91f2756a.jpg

Ground Truth Objects: 20 | Predicted Objects: 19

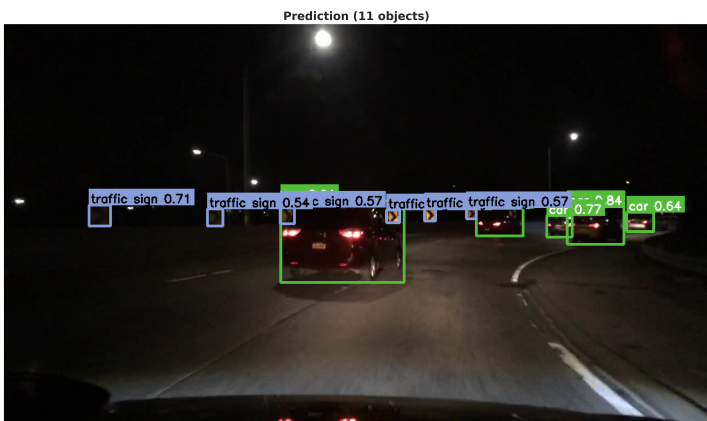
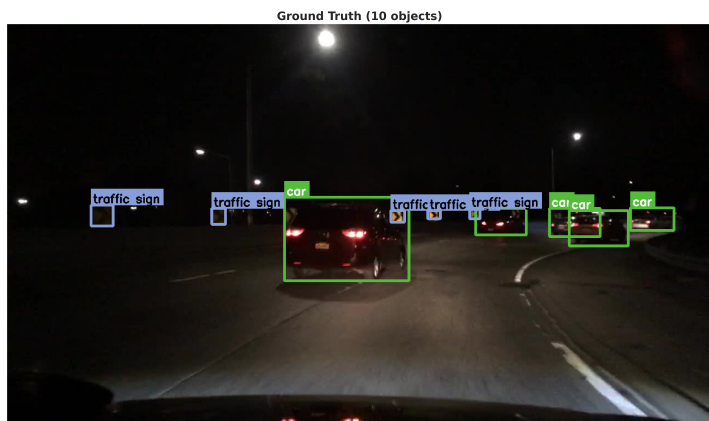


Sample #4: f8549008-f4b7323c.jpg

Ground Truth Objects: 11 | Predicted Objects: 10



Sample #5: edc30ef4-f0fa105a.jpg
 Ground Truth Objects: 10 | Predicted Objects: 11



Sample #6: dcf6de94-b7bf0ce4.jpg
 Ground Truth Objects: 20 | Predicted Objects: 15

